UNIT- II

PART A: Information Theory.

→ Information and entropy., Conditional and Joint Entropies

→ Mutual Information.

→ Information loss due to noise.

→ Types of channels and channel Capacity

→ Bandwidth - S/N trade off.

→ Hartley Shannon law.

PART B: Source Coding Techniques:

→ Need of Source Coding.

→ Various types of Source Codes.

→ Kraft Inequality Condition.

→ Entropy, Entropy Coding - Shannon Fano, Huffman Code

→ Source Coding to increase average information per bit

→ lossy Source Coding.

Information Theory.

→ The performance of the Communication s/m is measured interms of its error probability.

→ An error - less transmission is possible when probability of error at the receiver approaches zero.

⇒ Information theory is used for mathematical modeling and analysis of the Commⁿ s/ms.

Uncertainty:

Consider the source which emits the discrete symbols randomly from the set of fixed alphabet ie,

$$X = \{x_0, x_1, x_2 \ldots\ldots x_{K-1}\} \quad —(1)$$

→ The Various symbols in 'X' have probabilities of $P_0, P_1, P_2 \ldots$ etc, which can be written as

$$P(X = x_k) = P_k \quad \& \; k = 0,1,2 \ldots K-1 \quad —(2)$$

→ This set of probabilities satisfy the following condition.

$$\sum_{k=0}^{K-1} P_k = 1.$$

⇒ This is called discrete information source.

Consider the emission of symbol $X = x_k$ from sources

prob of x_k :

ie (i) $P_k = 0$, then such symbol is impossible.

(ii) $P_k = 1$, then such symbol is sure.

(iii) P_k = low, then there is more surprise (or) ~~[illegible]~~ Uncertainity.

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Definitation of Information:

→ Consider the Commn s/m, which transmits messages $m_1, m_2, m_3 \ldots$, with probabilities of occurances $P_1, P_2, P_3 \ldots$. The amount of information transmitted through the msg m_K with probability 'P_K' is given as.

Amt of Infn I_K, $I_K = \log_2\left[\frac{1}{P_K}\right]$

Units of information is '**bits**'

Properties:

i) If there is more uncertainty about the msg, infon carried is ~~also~~ more. high

ii) If receiver knows the msg being transmitted the amount of information carried is zero → $\boxed{I(x) = 0 \text{ for } p(x) = 1}$

iii) If I_1 is the information carried by msg m_1, I_2 carried by m_2, then the amount of infon carried due to m_1 & m_2 is '$I_1 + I_2$'

iv) If there are $M = 2^N$ equally likely messages, then amount of information carried by each msg will be 'N' bits.

Q: Calculate the amount of infn. If $P_K = 1/4$.

Soln = $I_K = \log_2[1/P_K] = \log_2\left[\frac{1}{1/4}\right] = \log_2\left[\frac{1}{1/2^2}\right] = \log_2 2^2 \Rightarrow 2\log_2 2$

$\Rightarrow 2$ bits

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Entropy: Average Information

→ Consider. M-different messages, $m_1, m_2, m_3 \cdots m_M$.

→ There probabilities of occurance as $P_1\ P_2\ P_3 \cdots P_M$.

→ Suppose that a sequence of 'L' messages are transmitted i.e,

$P_1 L$ messages of m_1 are transmitted lly.

$P_2 L \to m_2$; $P_3 L \to m_3$ .... $P_M L \to m_M$.

Hence infoⁿ due to massage m_1 will be,

$$I_1 = \log_2\left(\frac{1}{P_1}\right)$$

→ Since there are $P_1 L$ no of messages i.e, m_1. Then total infoⁿ due to all message of m_1 will be

$$I_1(\text{total}) = P_1 L \log_2(1/P_1)$$

lly. $P_2 L \to m_2$ $\quad I_2(\text{total}) = P_2 L \log_2(1/P_2)$ and so on.

→ Thus the total infoⁿ carried due to all sequences of 'L' messages will be

$$I(\text{total}) = I_1(\text{total}) + I_2(\text{total}) + \cdots + I_M(\text{total})$$

$$I(\text{total}) = P_1 L \log_2\left(\frac{1}{P_1}\right) + P_2 L \log_2\left(\frac{1}{P_2}\right) + P_3 L \log_2\left(\frac{1}{P_3}\right) \cdots P_M L \log_2\left(\frac{1}{P_M}\right)$$

$$\boxed{I(\text{total}) = \sum_{K=1}^{M} P_K L \log_2(1/P_K) \text{ bits}}$$

⇒ Average Info per message will be,

$$\text{Average info} = \frac{\text{Total info}}{\text{No of messages}} = \frac{I(\text{total})}{L}.$$

Average information is represented by Entropy. It is denoted by 'H'.

$$\therefore \text{Entropy } (H) = \frac{I(\text{total})}{L}$$

$$\therefore \text{Entropy} = \left[\sum_{k=1}^{M} L\, P_k \log_2(1/P_k)\right] / L$$

$$= \frac{L \sum_{k=1}^{M} P_k \log_2(1/P_k)}{L}$$

$$\boxed{\text{Entropy } (H) = \sum_{k=1}^{M} P_k \log_2\left(\frac{1}{P_k}\right)}$$

Units = bits/symbol

Properties:-

i) Entropy is zero, if the event is sure (or) impossible

i.e, $H = 0$ if $P_k = 0$ (or) 1.

ii) When $P_k = \frac{1}{M}$ for all the 'M' symbols, then the symbols are equally likely. For such source entropy is given as $\boxed{H = \log_2 M}$

iii) Upper bound on entropy is given as

$$\boxed{H_{max} = \log_2 M}$$

Information Rate: (R).

$\boxed{\text{Units} = \text{info bits/Second}}$

$$\text{Info rate } (R) = r \cdot H.$$

$H \rightarrow$ entropy

$r \rightarrow$ rate at which messages are generated.

msg, symbol, Info bits

$$R = [r \text{ in msg/sec}] \times [H \text{ is info bits/msg}]$$

Discrete Memory-less Channels:

⇒ Discrete memoryless channel has i/p 'x' & o/p 'y'; Both X & Y are random variables.

⇒ The channel is discrete when both X & Y are discrete.

⇒ The channel is called memoryless (zero memory) when current o/p depends only on current input.

⇒ The transition probability $P(y_j/x_i)$ is the conditional probability of y_j is received, for x_i transmission.

⇒ If $i=j$, then $P(y_j/x_i)$ represents a conditional probability of correct reception.

⇒ if $i \neq j$, then $P(y_j/x_i)$ represents a conditional probability of errors.

⇒ The transition probabilities of the channel can be represented by a matrix as follows:

$$P = \begin{bmatrix} P(y_1/x_1) & P(y_2/x_1) & \cdots & P(y_m/x_1) \\ P(y_1/x_2) & P(y_2/x_2) & \cdots & P(y_m/x_2) \\ P(y_1/x_3) & P(y_2/x_3) & \cdots & P(y_m/x_3) \\ \vdots & & & \\ P(y_1/x_n) & P(y_2/x_n) & \cdots & P(y_m/x_n) \end{bmatrix}$$

Each row/column of above matrix represents fixed i/p & fixed o/p.

⇒ The summation of all transition probabilities along the ~~tows~~ row is equal to 1. i.e,

$$p(y_1/x_1) + p(y_2/x_1) + p(y_3/x_1) + \cdots P(y_m/x_1) = 1.$$

⇒ This is applicable to other rows also. Hence we can write,

$$\sum_{j=1}^{m} p(y_j|x_i) = 1.$$

⇒ For the fixed i/p x_i, the o/p can be any of $y_1, y_2, y_3 \cdots y_m$.. The summation of all these probabilities is equal to 1.

⇒ From, probability theory, we know $P(AB) = P(B/A)P(A)$

→ $p(AB)$ is joint probability of A & B

⇒. Let $A = x_i$ & $B = y_j$, then $P(x_i, y_j) = p(y_j/x_i)\, p(x_i)$

⇒ If we add all the joint probabilities for fixed y_j, then we get $p(y_j)$ i.e,

$$\sum_{i=1}^{n} p(x_i, y_j) = p(y_j).$$

$$\therefore p(y_j) = \sum_{i=1}^{n} p(y_j/x_i) \cdot p(x_i)$$

Here $j = 1, 2 \ldots m$

→ Thus if i/p symbol & transition probabilities are given, then the calculation of o/p sym probabilities are possible.

→ Error will result when i^{th} symbol is transmitted but j^{th} symbol is received.

⇒ Hence error probability (P_e) is obtained as

$$P_e = \sum_{\substack{j=1 \\ j \neq i}}^{m} p(y_j).$$

→ Thus all the probabilities will contribute to error except $i=j$.

→ In case $i=j$, correct symbol will be received

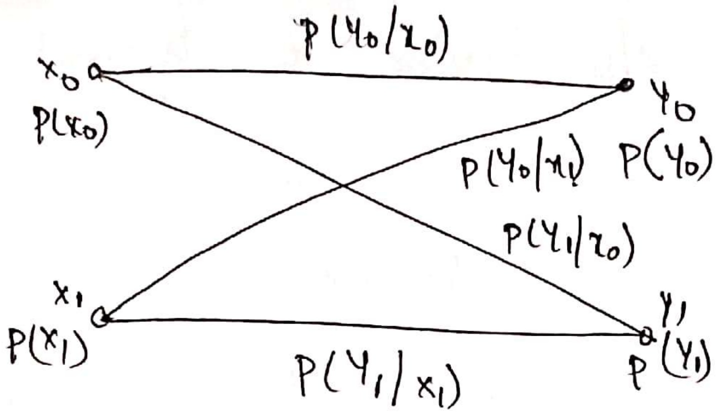
$$\therefore P_e = \sum_{\substack{j=1 \\ j \neq i}}^{m} \sum_{i=1}^{n} P(y_j/x_i)\, p(x_i)$$

→ The probability of correct reception will be

$$\boxed{P_c = 1 - P_e}$$

Binary Communication Channel

Consider the case of discrete channel when there are only two symbols transmitted.



→ Equations for probabilities of y_0 & y_1 as.

$$P(y_0) = P(y_0/x_0)\,P(x_0) + P(y_0/x_1)\cdot P(x_1).$$

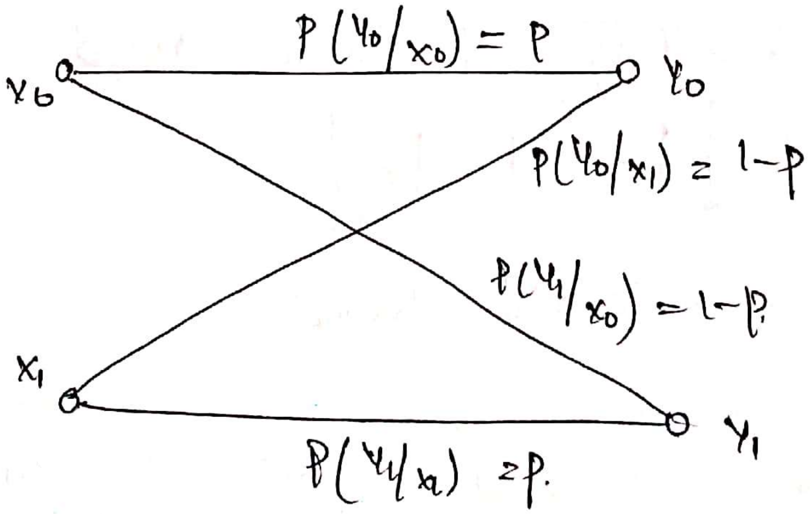
$$P(y_1) = P(y_1|x_1)\,P(x_1) + P(y_1/x_0)\,P(x_0).$$

⇒ Expressed in matrix form as.

$$\begin{bmatrix} P(y_0) \\ P(y_1) \end{bmatrix} = \begin{bmatrix} P(x_0) & P(x_1) \end{bmatrix} \begin{bmatrix} P(y_0/x_0) & P(y_1|x_0) \\ P(y_0/x_1) & P(y_1/x_1) \end{bmatrix}$$

Binary symmetric channel:

⇒ ~~By~~ A channel is said to be symmetric if $P(y_0/x_0) = P(y_1/x_1) = P$.



From the above channel, we can write,

$$\begin{bmatrix} p(y_0) \\ p(y_1) \end{bmatrix} = \begin{bmatrix} p(x_0) & p(x_1) \end{bmatrix} \begin{bmatrix} p & 1-p \\ 1-p & p \end{bmatrix}$$

Conditional Entropy: (or) Equivocation.

⇒ The Conditional entropy $H(X/Y)$ is called equivocation. It is defined as.

$$H(X/Y) = \sum_{i=1}^{M} \sum_{j=1}^{M} p(x_i, y_j) \log_2 \left[\frac{1}{p(x_i/y_j)} \right]$$

⇒ The Joint entropy $H(X,Y)$ is given as

$$H(X,Y) = \sum_{i=1}^{M} \sum_{j=1}^{M} p(x_i, y_j) \log_2 \frac{1}{p(x_i, y_j)}$$

⇒ The Conditional Entropy $H(X/Y)$ represents uncertainty of X, on avg, when Y is known.

Ily. Conditional Entropy $H(Y/X)$ represents uncertainty of Y on avg, when X is transmitted/known.

$$H(Y/X) = \sum_{i=1}^{M} \sum_{j=1}^{M} p(x_i, y_j) \log_2 \left[\frac{1}{p(y_j/x_i)} \right]$$

⇒ The Conditional entropy $H(X/Y)$ is an avg measure of uncertainty in X after 'Y' is received.

⇒ In otherwords $H(X/Y)$ represents the information lost in the noisy channel.

Rate of information Transmission over a Discrete channel:

→ The entropy of the symbol gives average amount of information going into the channel ie,

$$H(X) = \sum_{i=1}^{M} P_i \log_2 \left[\frac{1}{P_i}\right] \quad \text{—} (1)$$

⇒ We know that information rate $(R) = r \cdot H$.

$r \rightarrow$ rate at which symbols/messages are generated per second. (symbols/sec).

∴ The average rate of information going into the channel is given as.

$$D_{in} = r \cdot H(X) \text{ bits/sec.} \quad \text{—} (2)$$

⇒ But errors are introduced in the data during the txⁿ because of these errors, Some information is lost in the channel.

⇒ The Conditional entropy $H(X/y)$ is the measure of information lost in the channel. Hence the infoⁿ transmitted over the channel will be

$$\text{Txted info} = H(X) - H(X/y). \quad \text{—} (3)$$

⇒ Hence the average rate of information txⁿ (D_t) across the channel is taken as,

$$D_t = [H(X) - H(X/y)]\, r \text{ bits/sec.} \quad \text{—} (4)$$

→ The noise becomes very large, then X & Y (6) are statistically independent.

→ Then $H(X/Y) = H(X)$ & Hence no info^n is transmitted over the channel.

→ In case of ideal transmission $H(X/Y) = 0$, Hence.

$$\boxed{D_{in} = D_t} \quad \text{—— (5)}$$

i.e, i/p Info. rate is same as info^n rate across the channel. No info^n is lost when $H(X/Y) = 0$.

— x —

Capacity of a Discrete Memoryless channel:

The channel capacity is denoted by 'c', given as.

$$C = \max_{P(x)} \{D_t\},$$

Substitute eqn (4) in above exp^n,

$$C = \max_{P(x)} \{H(X) - H(X/Y)\} \cdot r.$$

→ The channel capacity can be defined as the max^m possible rate of info tx^n across the channel.

There are two theorems on channel capacity

1. Channel Coding theorem (Shannon 2nd theorem).
2. Shannon's Hartley for continuous channel.

Channel Coding Theorem:

Shannon's theorem states that it is possible to transmit information with an arbitarily small probability of error provided that information rate 'R' is less than or equal to a rate 'C' called channel capacity.

→ Thus channel capacity is the max^m info^n rate with which the error probability with in the tolerable limits.

STATEMENT:

Given a source of 'M' equally likely messages with $M >> 1$ which is generating info^n with rate 'R'. Given channel with channel capacity 'C', then if $\boxed{R \leq C}$ there exist a coding technique such that the o/p of the source may be transmitted over the channel with a probability of error in the rec'd msg, which may be made arbitarily small.

Shannon's Hartley for Continuous channel Capacity Theorem:

⇒ Shannon theorem of channel Capacity is added to the channel in which the noise is gaussian is known as shannon's Hartley theorem.

⇒ Its also known as ① Info Capacity theorem.
② Shannon's Hartley theorem for Gaussian channels
③ Negative statement of channel Coding Theorem $\boxed{R > C}$

⇒ The channel capacity of wide band limited gaussian channel is.

$$C = B \log_2 \left[1 + \frac{S}{N}\right] \text{ bits/sec.}$$

Where. B = Band width, C = Channel capacity
S/N = SNR, S = signal power.
N ⇒ Noise power

⇒ signal power is given by.

$$\text{power 'p'} = \int_{-B}^{B} \text{power spectral density}$$

⇒ PSD of white Noise = $N_0/2$

⇒ Noise power 'N' $= \int_{-B}^{B} N_0/2 \, df$ ⇒ $\frac{N_0}{2} \int_{-B}^{B} df$

⇒ $\frac{N_0}{2} [f]_{-B}^{B}$ ⇒ $\frac{N_0}{\cancel{2}} [\cancel{2}B]$ ⇒ $\boxed{N = N_0 B}$

Trade of B/w Bandwidth + SNR:

⇒ Channel capacity of gaussian channel is given by

$$C = B \log_2 \left[1 + \frac{S}{N}\right] \text{ bits/sec.} \quad \text{— (A)}$$

⇒ ∴ channel capacity 'C' depends on B/W & SNR.

Channel capacity at (1) Infinite Capacity
(2) Limited Capacity.

Channel with '∞' Capacity:

⇒ If there is no noise in channel i.e, $N=0$.

Hence $\frac{S}{N} = '\infty'$. Such channel is called Noiseless Channel.

$$C = B \log_2 [1 + \infty]$$

$$\boxed{C = \infty}$$

We can state that Noiseless Channels has '∞' Capacity.

Infinite BW channel has limited capacity

(or)

Channel with limited capacity

As we know, $N = N_0 \cdot B$.

If $B = \infty \Rightarrow N = N_0(\infty)$

$$\boxed{N = \infty}$$

⇒ As BW ↑ Noise power ↑, So S/N ↓.

⇒ Therefore channel capacity depends on SNR.

⇒ Even if B approaches '∞', capacity doesnot approach infinity.

⇒ As $B \to \infty$, capacity approaches an upper limit, which is given by.

$$C_\infty = \lim_{B \to \infty} C = 1.44 \, S/N.$$

Proof:

→ $N = N_0 \cdot B$

Substitute in eq (A)

$$C = B \log_2 \left[1 + \frac{S}{N_0 B} \right] \text{ bits/sec.}$$

Arrange the above eqn as.

$$C = \frac{S}{N_0} \cdot \frac{N_0 B}{S} \log_2 \left[1 + \frac{S}{N_0 \cdot B} \right]$$

$$= \frac{S}{N_0} \log_2 \left[1 + \frac{S}{N_0 \cdot B} \right] \left(\frac{N_0 B}{S} \right)$$

$$C \Rightarrow \frac{S}{N_o} \log_2 \left[1 + \frac{S}{N_o \cdot B}\right]^{\frac{1}{S/N_o \cdot B}}$$

Apply limits as $B \to \infty$.

$$C_\infty = \lim_{B \to \infty} \frac{S}{N_o} \log_2 \left[1 + \frac{S}{N_o \cdot B}\right]^{1/(S/N_o \cdot B)}$$

$$C_\infty = \lim_{x \to 0} \frac{S}{N_o} \log_2 (1 + x)^{1/x}$$

Assume $\frac{S}{N_o \cdot B} = x$, $B \to \infty$, $x = 0$.

We know that, $\lim_{x \to \infty} (1+x)^{1/x} = e$,

$$\therefore C_\infty = \lim_{x \to 0} \frac{S}{N_o} \log_2 e.$$

$$\boxed{C_\infty = 1.442 \frac{S}{N_o}}$$

This eqn gives the upper limit on channel Capacity as BW approaches ∞.

$\frac{S}{N}$, $\frac{E_b}{N_o}$ trade off, Rate/BW:

Let the s/m is transmitting at the rate 'R', which is equal to channel Capacity 'c', then avg transmitted s/g power will be ($E_b \to$ energy/bit)

$$S = E_b \cdot C$$

→ Capacity of Continuous channel is given as.

eqn (A) $\Rightarrow C = B\log_2\left[1+\frac{S}{N}\right]$.

$\therefore \quad S = E_b \cdot C$

$N = N_0 \cdot B$

$$\therefore \quad C = B\log_2\left[1+\frac{E_b \cdot C}{N_0 \cdot B}\right]$$

$$\Rightarrow \quad \frac{C}{B} = \log_2\left[1+\frac{E_b \cdot C}{N_0 \cdot B}\right]$$

$$\frac{C}{B} = \frac{\log_e\left[1+\frac{E_b \cdot C}{N_0 \cdot B}\right]}{\log_e 2}$$

$$\log_e 2 \cdot \left(\frac{C}{B}\right) = \log_e\left[1+\frac{E_b \cdot C}{N_0 \cdot B}\right]$$

$$\cancel{0.693}.\left(\frac{C}{B}\right)^{\log_e 2.} = \log_e\left[1+\frac{E_b \cdot C}{N_0 \cdot B}\right]$$

$$\left[a\log_e 2 \Rightarrow \log_e 2^a\right]$$

$$\log_e 2^{(C/B)} = \log_e\left[1+\frac{E_b \cdot C}{N_0 \cdot B}\right]$$

$$2^{C/B} = \left[1+\frac{E_b \cdot C}{N_0 \cdot B}\right]$$

$$\Rightarrow 2^{C/B} - 1 = \frac{E_b \cdot C}{N_0 \cdot B}.$$

$$-1 + 2^{C/B} = \left(\frac{E_b}{N_0}\right)\left(\frac{C}{B}\right)$$

$$\Rightarrow \boxed{\frac{2^{C/B}-1}{(C/B)} = \frac{E_b}{N_0}}$$

In this eqn, $\frac{E_b}{N_0}$ is ratio of energy/bit to Noise Spectral density.

$\frac{C}{B}$ is Bandwidth efficiency.

When $C = R_b$. then $\frac{E_b}{N_0} = \frac{2^{(R_b/B)} - 1}{(R_b/B)}$

$R_b \rightarrow$ infon rate per bit

$\left(\frac{R_b}{B}\right)$ - Rate BW

But $C_\infty = 1.44\ S/N_0$.

Hence $S = E_b \cdot C \Rightarrow E_b \cdot C_\infty$, Subsitute $S = E_b C_\infty$ in C_∞

$\therefore C_\infty = 1.44 \frac{E_b C_\infty}{N_0}$

$\frac{\cancel{C_\infty}}{1.44} = \frac{E_b \cdot \cancel{C_\infty}}{N_0}$

$\Rightarrow \frac{E_b}{N_0} = \frac{1}{1.44}$

$$\boxed{\frac{E_b}{N_0} = 0.693}$$

$\therefore \frac{S}{N}$ in dB is taken as

$\left(\frac{E_b}{N_0}\right)_{dB} = 10 \log_{10} \frac{E_b}{N_0} = 10 \log_{10}(0.693)$

$= -1.6\,dB$

$$\boxed{\left(\frac{E_b}{N_0}\right)_{dB} = -1.6\,dB.}$$ at $B \rightarrow \infty$

$\therefore \frac{E_b}{N_0}$ Value is called shannon limit.

$\Rightarrow$ capacity at shannon's limit is $C_\infty = 1.44 \frac{S}{N_0}$.

For $R_b < C \rightarrow$ Error free Tx^n is possible.

$R_b > C \rightarrow$ Noise ~~cor~~ Currupted Tx^n

$R_b = C \rightarrow$ Capacity boundary.

Probability of error for an ideal s/m is written as

$$P_e = \begin{cases} 1 & \text{for } R \geq C \\ 0 & \text{for } R < C. \end{cases}$$

$P(x_i/y_j) \rightarrow$ Conditional prob. $x_i \rightarrow$ tx'd & y_j isRxd

$P(x_i) \rightarrow$ prob of x_i symbols for tx'n.

Mutual Information

Defined as:

The amount of information transferred when x_i is tx'd and y_j is received.

Represented by $I(x_i, y_j) \rightarrow$ mutual Inf^n.

Given as $I(x_i, y_j) = \log_2 \dfrac{P(x_i/y_i)}{P(x_i)}$ bits.

$I(x|y)$ or $I(X;Y)$ is Avg mutual Info. $\quad I(x_i, y_i) \Rightarrow I(X;Y)$

$$\boxed{I(X;Y) = H(X) - H(X|Y)} \text{ bits / symbol.}$$

sym/msg.

$H(X) \rightarrow$ uncertainity about channel i/p ~~observed~~ before ~~o/p~~ channel o/p is observed

$H(X|Y) \rightarrow$ uncertainity about channel i/p after the channel o/p is observed.

$I(X;Y) \rightarrow$ Uncertainity about the channel i/p that is resolved by observing channel o/p.

Avg mutual Info $[I(x;y)]$ is defined as the amount of Source info^n gained per Received Symbol..

$$\boxed{I(x;y) = \sum_{i=1}^{n} \sum_{j=1}^{m} P(x_i, y_j)\, I(x_i, y_j)}$$

Properties of Mutual Info.

(1) $I(x;y) = I(y;x)$.

ie, Mutual Inf^n of the channel is symmetric.

(2) $I(x;y) \geq 0$; MI is always positive.

(3) $I(x;y) = H(x) - H(x/y)$

$H(y) - H(y/x)$

MI can be expressed interms of entropies of channel i/p (a) o/p. & Conditional entropies.

(4) $I(x;y) = H(x) + H(y) - H(x,y)$.

MI is related to the joint entropy. $H(x;y)$

— x —.

Properties of Mutual Information:

(i) The mutual information of the channel is symmetric i.e,

$$I(X;Y) = I(Y;X)$$

(ii) The mutual information can be expressed in terms of entropies of channel input or o/p of conditional entropies

i.e, $$I(X;Y) = H(X) - H(X/Y)$$
$$= H(Y) - H(Y/X)$$

(iii) The mutual information is always positive, i.e,

$$I(X;Y) \geq 0$$

(iv) The mutual information is related to the joint entropy $H(X;Y)$ by following relation:

$$I(X;Y) = H(X) + H(Y) - H(X,Y)$$

Problems:

1. A source transmits two independent messages with probabilities of 'P' and (1-P) respectively. Prove that the entropy is maximum when both the messages are equally likely. Plot the variation of entropy (H) as a function of probability 'P' of the messages.

Sol:- Entropy is given as,

$$H = \sum_{K=1}^{M} P_K \log_2\left[\frac{1}{P_K}\right]$$

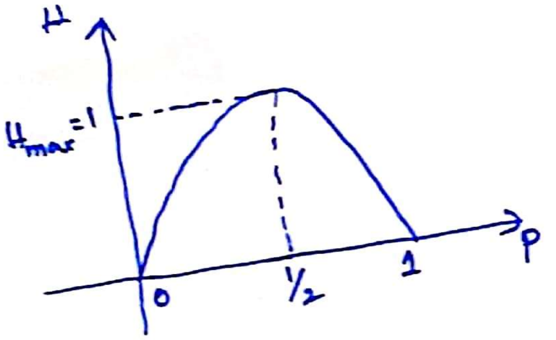
for two messages above equation will be,

$$H = \sum_{K=1}^{2} P_K \log_2\left[\frac{1}{P_K}\right]$$

$$= P_1 \log_2\left[\frac{1}{P_1}\right] + P_2 \log_2\left[\frac{1}{P_2}\right]$$

Here we have two messages with probabilities $P_1 = P$ and $P_2 = 1-P$.

$$\therefore \; H = P \log_2\left(\frac{1}{P}\right) + (1-P)\log_2\left[\frac{1}{1-P}\right]$$



P	H
0	0
0.2	0.7215
0.4	0.9709
0.5	1
0.6	0.9709
0.8	0.7215
1	0

→ The maximum of 'H' occurs at $P = \frac{1}{2}$, i.e when messages are equally likely

→ Putting $P = \frac{1}{2}$. in above eqn we get,

$$H_{max} = \frac{1}{2}\log_2(2) + \frac{1}{2}\log_2(2)$$

$$= \log_2(2) = \frac{\log_{10}(2)}{\log_{10}(2)} = 1 \text{ bit/message}$$

2. For a discrete memoryless source there are three symbols with probabilities $P_1 = \alpha$ and $P_2 = P_3$. Determine the entropy of the source and sketch its variation for different values of α.

Sol:- The three probabilities are,

$$P_1 = \alpha \quad \& \quad P_2 = P_3$$

we know $P_1 + P_2 + P_3 = 1$

$\alpha + P_2 + P_2 = 1$ since $P_2 = P_3$

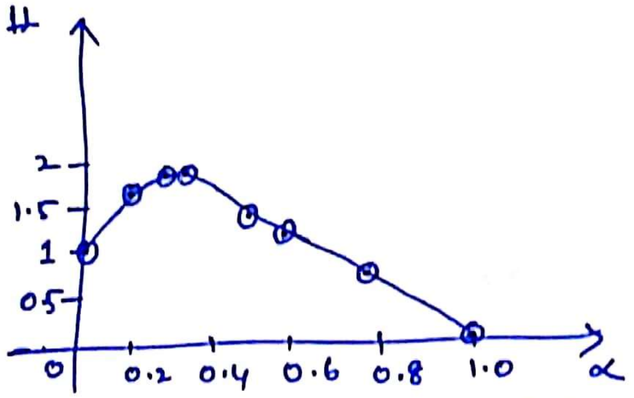
$$\therefore \quad P_2 = \frac{1-\alpha}{2}$$

$$P_3 = P_2 = \frac{1-\alpha}{2}$$

Entropy $H = \sum_{K=1}^{3} P_K \log_2 \frac{1}{P_K}$

$$= \alpha \log_2\left(\frac{1}{\alpha}\right) + \left(\frac{1-\alpha}{2}\right) \log_2\left(\frac{2}{1-\alpha}\right) + \left(\frac{1-\alpha}{2}\right) \log_2\left(\frac{2}{1-\alpha}\right)$$

$$= \alpha \log_2\left(\frac{1}{\alpha}\right) + (1-\alpha) \log_2\left(\frac{2}{1-\alpha}\right)$$



α	H
0	1
0.2	1.52
0.3	1.58
0.4	1.57
0.5	1.5
0.6	1.37
0.8	0.92
1.0	0

③ An analog signal is bandlimited to B Hz and sampled at Nyquist rate. The samples are quantized into 4 levels. Each level represents one message. Thus there are 4 messages. The probabilities of occurance of these 4 levels (messages) are $P_1 = P_4 = \frac{1}{8}$ & $P_2 = P_3 = \frac{3}{8}$. Find out information rate of the source.

Sol:- (i) To calculate entropy (H):

$$H = P_1 \log_2\left(\frac{1}{P_1}\right) + P_2 \log_2\left(\frac{1}{P_2}\right) + P_3 \log_2\left(\frac{1}{P_3}\right) + P_4 \log_2\left(\frac{1}{P_4}\right)$$

$$= \frac{1}{8}\log_2(8) + \frac{3}{8}\log_2\left(\frac{8}{3}\right) + \frac{3}{8}\log_2\left(\frac{8}{3}\right) + \frac{1}{8}\log_2(8)$$

$$H = 1.8 \text{ bits/message}$$

(ii) To calculate message rate (r):

We know signal is sampled at nyquist rate.

Nyquist rate for B Hz bandlimited signal is,

$$\text{Nyquist rate} = 2B \text{ samples/sec.}$$

Since every sample generates one message signal,

Messages per second, $r = 2B$ messages/sec.

(iii) To calculate information rate (R):

$$R = rH$$

$$R = 2B \text{ messages/sec} \times 1.8 \text{ bits/message}$$

$$= 3.6\,B \text{ bits/sec.}$$

4) consider a telegraph source having two symbols dot and dash. The dot duration is 0.2 sec; and dash duration is 3 times of dot duration. The probability of the dot's occurring is twice that of dash, and the time between symbols is 0.2 seconds. Calculate information rate of telegraph source.

Sol:- (i) To calculate probabilities of dot and dash:

Let the probability of dash be 'p'. Then probability of dot will be '2p'.

$$p + 2p = 1$$

$$\therefore p = 1/3$$

Thus $P(\text{dash}) = \frac{1}{3}$ & $P(\text{dot}) = \frac{2}{3}$

(ii) To calculate entropy of the source:

$$H = \sum_{K=1}^{M} P_K \log_2\left(\frac{1}{P_K}\right)$$

for dots and dash, entropy will be

$$H = \frac{1}{3}\log_2(3) + \frac{2}{3}\log_2\left(\frac{3}{2}\right)$$

$$= 0.9183 \text{ bits/symbol.}$$

(iii) To calculate average symbol rate:

Given, dot duration $= T_{dot} = 0.2\,sec$

dash duration $= T_{dash} = 3 \times 0.2 = 0.6\,sec.$

Duration between symbols = $T_{symbols}$ = 0.2 sec.

Let us consider string of 1200 symbols.

Number of dots = $1200 \times P(\text{dots}) = 1200 \times \frac{2}{3} = 800$

No. of dash = $1200 \times P_{(\text{dash})} = 1200 \times \frac{1}{3} = 400$

Total time for this string is

T = dots duration + dash duration + (1200 × time between symbols)

$= (800 \times 0.2) + (400 \times 0.6) + (1200 \times 0.2)$

$= 640$ sec.

Average symbol rate is $r = \frac{1200}{T} = \frac{1200}{640} = 1.875$ symbols/sec

(iv) To calculate information rate:

$R = rH = 1.875 \times 0.9183$

$= 1.7218$ bits/sec.

⑤ Prove that $H(X,Y) = H(X/Y) + H(Y)$

$= H(Y/X) + H(X)$

Sol:- $H(X,Y) = \sum_{i=1}^{M} \sum_{j=1}^{M} P(x_i, y_j) \log_2 \left(\frac{1}{P(x_i, y_j)} \right)$

$= -\sum_{i=1}^{M} \sum_{j=1}^{M} P(x_i, y_j) \log_2 P(x_i, y_j)$

from probability theory,

$$P(x_i, y_j) = P(x_i/y_j)\,P(y_j)$$

$$\therefore H(X,Y) = -\sum_{i=1}^{M}\sum_{j=1}^{M} P(x_i, y_j)\log_2\left[P(x_i/y_j)\,P(y_j)\right]$$

we know $\log_2\left[P(x_i/y_j)\,P(y_j)\right] = \log_2 P(x_i/y_j) + \log_2 P(y_j)$

Hence above Eqn becomes,

$$H(X,Y) = -\sum_{i=1}^{M}\sum_{j=1}^{M} P(x_i, y_j)\log_2 P(x_i/y_j)$$

$$-\sum_{i=1}^{M}\sum_{j=1}^{M} P(x_i, y_j)\log_2 P(y_j)$$

$$= \sum_{i=1}^{M}\sum_{j=1}^{M} P(x_i, y_j)\log_2\left\{\frac{1}{P(x_i/y_j)}\right\}$$

$$-\sum_{j=1}^{M}\left\{\sum_{i=1}^{M} P(x_i, y_j)\right\}\log_2 P(y_j)$$

$\rightarrow \sum_{i=1}^{M} P(x_i, y_j) = P(y_j)$

$$\therefore H(X,Y) = H(X/Y) - \sum_{j=1}^{M} P(y_j)\log_2 P(y_j)$$

$$= H(X/Y) + \sum_{j=1}^{M} P(y_j)\log_2\left\{\frac{1}{P(y_j)}\right\}$$

$$\therefore H(X,Y) = H(X/Y) + H(Y)$$

thus the first given equation is proved.

From the probability theory $P(AB) = P(B/A) \cdot P(A)$

$$\therefore P(x_i, y_j) = P(y_j/x_i)\, P(x_i)$$

Put this in $H(X,Y) = -\sum_{i=1}^{M} \sum_{j=1}^{M} P(x_i, y_j) \log_2 \left[P(y_j/x_i) P(x_i) \right]$

$$= -\sum_{i=1}^{M} \sum_{j=1}^{M} P(x_i, y_j) \log_2 P(y_j/x_i)$$

$$- \sum_{i=1}^{M} \sum_{j=1}^{M} P(x_i, y_j) \log_2 P(x_i)$$

$$= \sum_{i=1}^{M} \sum_{j=1}^{M} P(x_i, y_j) \log_2 \left[\frac{1}{P(y_j/x_i)} \right]$$

$$- \sum_{i=1}^{M} \left\{ \sum_{j=1}^{M} P(x_i, y_j) \right\} \log_2 P(x_i)$$

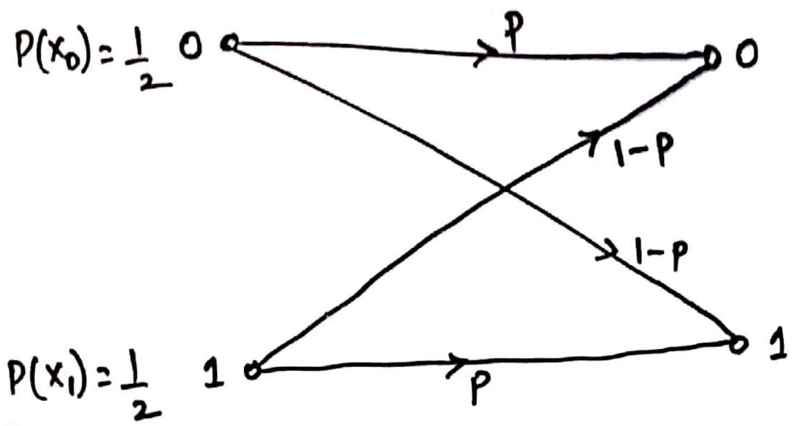
$$\Rightarrow \sum_{j=1}^{M} P(x_i, y_j) = P(x_i)$$

$$\therefore H(X,Y) = H(Y/X) - \sum_{i=1}^{M} P(x_i) \log_2 P(x_i)$$

$$= H(Y/X) + \sum_{i=1}^{M} P(x_i) \log_2 \left[\frac{1}{P(x_i)} \right]$$

$$\therefore H(X,Y) = H(Y/X) + H(X)$$

(6)



Find the rate of information transmission across this channel for $P=0.8$ & 0.6. The symbols are generated at the rate of 1000 per second. $P(x_0) = P(x_1) = \frac{1}{2}$. Also determine channel input information rate.

Sol:- (i) To obtain entropy of the source:-

$$H(x) = P(x_0)\log_2\left[\frac{1}{P(x_0)}\right] + P(x_1)\log_2\left[\frac{1}{P(x_1)}\right]$$

$$= \frac{1}{2}\log_2(2) + \frac{1}{2}\log_2(2)$$

$$= 1 \text{ bit/symbol.}$$

(ii) To obtain input Information rate:-

$$D_{in} = r\,H(x)$$

$$r = 1000 \text{ symbols/sec}$$

$$D_{in} = 1000 \times 1 = 1000 \text{ bits/sec.}$$

(iii) To obtain $P(y_0)$ & $P(y_1)$:

$$\begin{bmatrix} P(y_0) \\ P(y_1) \end{bmatrix} = \begin{bmatrix} P(x_0) & P(x_1) \end{bmatrix}_{1\times 2} \begin{bmatrix} P & 1-P \\ 1-P & P \end{bmatrix}_{2\times 2}$$

$$= \begin{bmatrix} \frac{1}{2} & \frac{1}{2} \end{bmatrix} \begin{bmatrix} P & 1-P \\ 1-P & P \end{bmatrix}$$

$$= \begin{bmatrix} \frac{1}{2}P + \frac{1}{2}(1-P) \\ \frac{1}{2}(1-P) + \frac{1}{2}P \end{bmatrix} = \begin{bmatrix} \frac{1}{2} \\ \frac{1}{2} \end{bmatrix}$$

$\therefore$ $P(y_0) = 1/2$ & $P(y_1) = 1/2$.

(iv) To obtain $P(x_i, y_j)$ and $P(x_i/y_j)$:

$\rightarrow$ $P(x_0\ y_0) = P(y_0|x_0) \cdot P(x_0)$

$= P \times \frac{1}{2} = \frac{P}{2}$

$\rightarrow$ $P(x_1\ y_0) = P(y_0|x_1) \cdot P(x_1)$

$= (1-P) \times \frac{1}{2} = \frac{(1-P)}{2}$

$\rightarrow$ $P(x_0\ y_1) = P(y_1|x_0) \cdot P(x_0)$

$= (1-P) \times \frac{1}{2} = \frac{(1-P)}{2}$

$\rightarrow$ $P(x_1\ y_1) = P(y_1|x_1) \cdot P(x_1)$

$= P \times \frac{1}{2} = \frac{P}{2}$

$$P(x_0/y_0) \cdot P(y_0) = P(y_0/x_0) \cdot P(x_0)$$

$\rightarrow$ $$P(x_0/y_0) = \frac{P(y_0/x_0) \cdot P(x_0)}{P(y_0)}$$

$$= \frac{P \times \frac{1}{2}}{\frac{1}{2}} = P$$

$\rightarrow$ $$P(x_1/y_0) = \frac{P(y_0/x_1) \cdot P(x_1)}{P(y_0)}$$

$$= \frac{(1-P) \times \frac{1}{2}}{\frac{1}{2}} = (1-P)$$

$\rightarrow$ $$P(x_0/y_1) = \frac{P(y_1/x_0) \cdot P(x_0)}{P(y_1)} = \frac{(1-P) \times \frac{1}{2}}{\frac{1}{2}} = (1-P)$$

$\rightarrow$ $$P(x_1/y_1) = \frac{P(y_1/x_1)\, P(x_1)}{P(y_1)} = \frac{P \times \frac{1}{2}}{\frac{1}{2}} = P$$

(v) To obtain information rate across the channel:

The conditional entropy $H(X/Y)$ is given as.

$$H(X/Y) = \sum_{i=1}^{M} \sum_{j=1}^{M} P(x_i, y_j) \log_2 \left[\frac{1}{P(x_i, y_j)}\right]$$

Here $M = 2$

$$H(X/Y) = P(x_0, y_0) \cdot \log_2\left[\frac{1}{P(x_0/y_0)}\right] + P(x_0, y_1) \cdot \log_2\left[\frac{1}{P(x_0/y_1)}\right]$$

$$+ P(x_1, y_0) \log_2\left[\frac{1}{P(x_1/y_0)}\right] + P(x_1, y_1) \log_2\left[\frac{1}{P(x_1/y_1)}\right]$$

$$H(X/Y) = \frac{1}{2} P \log_2\left(\frac{1}{P}\right) + \frac{1}{2}(1-P) \log_2\left[\frac{1}{1-P}\right] + \frac{1}{2}(1-P) \log_2\left[\frac{1}{1-P}\right]$$

$$+ \frac{1}{2} P \log_2\left(\frac{1}{P}\right)$$

$$H(X/Y) = P \log_2\left(\frac{1}{P}\right) + (1-P) \log_2\left[\frac{1}{1-P}\right]$$

→ For $P = 0.8$, $H(X/Y) = 0.8 \log_2\left(\frac{1}{0.8}\right] + (1-0.8) \log_2\left[\frac{1}{(1-0.8)}\right]$

$= 0.721928$ bits/symbol.

→ Information transmission rate across the channel will be,

$D_t = \{H(x) - H(X/Y)\} \times r$

$D_t = (1 - 0.721928) \times 1000 = 278$ bits/sec.

→ For $P = 0.6$, $H(X/Y) = 0.6 \log_2\left[\frac{1}{0.6}\right] + (1-0.6) \log_2\left[\frac{1}{(1-0.6)}\right]$

$= 0.97$ bits/symbol.

$\therefore D_t = (1 - 0.97) \times 1000 = 29$ bits/sec.

→ This shows that the information transmission rate decreases rapidly as 'p' approaches $\frac{1}{2}$.